

Assignment 2(a): Git & GitHub Viva (Detailed Q&A;)

BASIC LEVEL

Q1: What is Git?

Git is a distributed version control system used to track changes in source code during development.

Q2: What is GitHub?

GitHub is a cloud-based platform used to host and manage Git repositories.

Q3: What is a repository?

A repository is a storage location where project files and their history are maintained.

Q4: What is a commit?

A commit is a snapshot of changes made to files in the repository.

Q5: What is staging area?

It is a temporary area where changes are prepared before committing.

INTERMEDIATE LEVEL

Q6: What is a branch?

A branch is an independent line of development used to work on features separately.

Q7: What is origin?

Origin is the default name given to the remote repository.

Q8: What is git push?

It uploads local commits to the remote repository.

Q9: What is git pull?

It fetches and merges changes from remote repository to local system.

Q10: What is git clone?

It creates a local copy of a remote repository.

ADVANCED LEVEL

Q11: What is distributed version control?

Every user has a complete copy of the repository including history.

Q12: Difference between pull and fetch?

Pull = fetch + merge, Fetch = only downloads changes.

Q13: What is merge conflict?

Occurs when Git cannot automatically merge changes.

Q14: What is HEAD?

HEAD is a pointer to the latest commit.

Q15: What is .git folder?

It stores all repository metadata and history.

IMPORTANT EXAM QUESTIONS

Q16: Explain Git workflow.

Initialize repository → Add files → Commit changes → Connect to remote → Push to GitHub.

Q17: Why commit message is important?

It helps understand what changes were made.

Q18: Why push error occurs?

Due to mismatch between local and remote repository histories.

Q19: Advantages of Git?

Version tracking, collaboration, backup, branching.

Q20: Difference between local and remote repo?

Local is on system, remote is hosted on GitHub.